Determining hysteresis, remanence and coercive fields of a ferromagnet

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Kurzbeschreibung des Inhalts unter Angabe der Methodik, der Zielsetzung und der wichtigsten Ergebnisse

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1 Introduction

Ferromagnets are very important nowadays as they have a memory contrary to para- and diamagnets. Thus they are used for electromotors of cars or other machines, CDs or harddiscs. They are utilized as datastore because of their characteristic hysteresis, which does not follow a function of B(H) as it depends on the state it was in before (magnetised or not) and the magnetic field, which effects the ferromagnet. Sizes like remanence, coercive fields and saturation magnetisation are specific for the material being used. Hence we determine sizes like those by hysteresis curves in the following paper.

2 Theory

First of all it is important to clear some terms and definitions in relation to the theory of these following experiments. (As before mentioned ferromagnets are used to save information in long term like CD´s. But there are also other types of magnets, which can´t be used for saving data for a longer time period. To compare some attributes of ferromagnets and one of the other formes named diamagnets, wood is used as an example of diamagnetism. Diamagnets, other than ferromagnets are not able to construct a magnetic field without an outer magnetic induction. They are unmagnetic. So if there is an exterior magnetic field which tries to put information inside the diamagnet, all of these pieces of information are gone when the outer field is turned off because of the material structure.) Ferromagnets as mentioned before are able to build their own magnetic field without an exterior magnetic induction. That´s because of the material structure of ferromagnets. The magnetic moments on the inside of an atom tend to align parallel. So if there is an outer magentic field the moments inside the magnet orient themselves towards the direction of the outside one. xxxxxxxxxxxxxxxxxxxxxxx This happens in a macroscopic scale. In a minor dimension,the microscopic one, so called magnetic domains direction themselves. Magnetic domains are regions in a magnetic material where the magnetization points in the same direction. These domains get bigger and their borders migrate if the outside magnetic field gets stronger, so more and more magnetic moments direction themselves like the tendency of the outer field. These borders are called domain walls. If there is no outside influence there are different materialspecific magnetic domains in one substance and the domain walls are steady. So if the outer magnetic field H increases and therefore the domain walls expand, the magnetic flux B also increases. This relation B(H) is shown in so called hysteresis curves. Regarding to this curves there are a few terms, which should be explained beforehand. xxx The highest and lowest point of the curve is called the saturation magnetization. xxx From this special point all magnetic domains are directioned parallel to the outer field. With hitting this spot the magnetic flux B is only increasing with the incline of the vacuum permeability μ_0. If now the outer magnetic field is reduced constantly some of the magnetic domains start to change their orientation because of thermal fluctations and the magnetic flux decreases. This happens on a different trajectory than before because the thermal fluctation is less than the spin interaction in ferromagnetic materials with roomtemperatur. So if the point $H = 0$ is passed there is magnetism left, which is called remanence. After that the direction of the outer magnetic field is in-

verted and enlarged. Eventually the curve gets to the spot where the magnetic flux is completely gone. The value of H at this point is called Coercivity. Now the procedure is the same like before but with a reversed magnetic field. The hysteresis curve is complete if the saturation magnetization from the beginning is achieved again. So every hysteresis curve is point-symmetrical. When the magnetic domains shift in a different direction during this curve energy in form of heat gets lost. This heat equals the area inside the hysteresis curve and can be calculated with the formula:

$$\xi = \frac{E}{m} = \frac{1}{\rho} \oint BdH \tag{1}$$

3 Experiment

In the following experiments hysteresis curves of different materials were taken with the experiment set-up given in figure XX. Also with the use of these figures some parameters like remanence or coercitivity could be identified. The installation consists of two Toroidal inductors, one with the material to be analysed. One generates the outer magnetic field which induces a voltage and electricity in the induction coil. The generated magnetic field H is given by the voltage drop U_R measured with a known resistance R with the formula

$$H = \frac{N_1 * U_R}{\pi * \frac{1}{2} * (d_i + d_a) * R} \tag{2}$$

with N_1 the winding number, d_i and d_a the inner and outer diameter of the field creating inductor. A resistance with the value $R = (66,7 \pm 1,4)\Omega$ is used for measuring low electricity . Whereas for higher electricity the measuring resistor $R_m = 0,20\ \Omega$ is utilized. With the change of the flux density a voltage surge in the induction coil is created. Via the built in fluxmeter, this voltage surge can be measured against time to get a tension which is proportional to the flux density B calculated with the formula:

$$B = \frac{\gamma_f * U_f}{N_2 * A_{eff}} \tag{3}$$

Where γ_f is an adjustable proportionality factor of U_f. N_2 is the winding number of the induction coil. A_{eff} the area which is actual saturated by the magnetic field is measured with

$$A_{eff} = \frac{1}{2} * (d_a - d_i) * h * f \tag{4}$$

with the diameters, the hight and the bulk factor of the core. With all that it is possible to plot the hysteresis curves with the Interface called PASCO Universal Interface 550 Analog-Digital-Wandler. However because of the functionality of the fluxmeter and the integrated operational amplifiers all values are shifted in the direction of the flux density. To get the right plots these data points have to be centered with:

$$U_c = U_B - \frac{1}{2}(U_{min} + U_{max}) \tag{5}$$

Whereas U_c is the centered voltage of the induction coil and U_B the induced voltage in it. U_{min} and U_{max} are the lowest and highest measured voltages. Because there is no infinite input resistence plugged in the fluxmeter there is also a linear shift, rising by time, regarding to the data points of B. How this disruptive factor is adjusted is explained in xxxx on the basis of the hysteresis curve of wood.

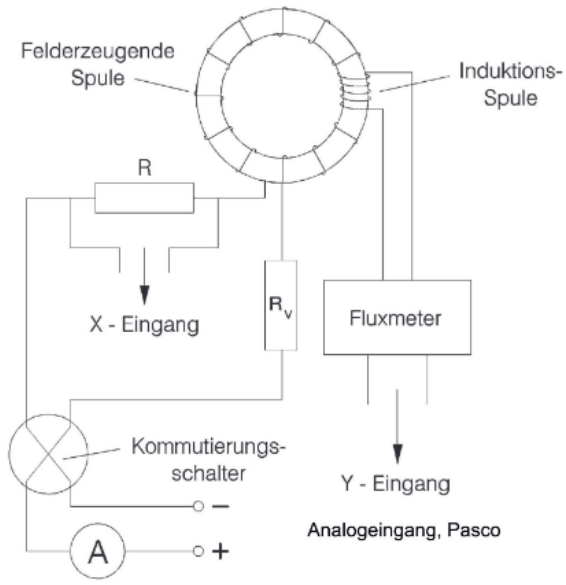

Abb. 1

3.1 Measurement with the core PERMENORM 5000H2

In this first experiment the hysteresis curve of the core PERMENORM 5000H2 is plotted. The material has an iron-nickel alloying. The resistance $R = 66,7\Omega$ is used with $I_{max} = 100$mA of electricity. The procedure to get the curve is described in the headline 3. The received curve is not closed. This is up to the disruptive factors, but can be corrected afterwards. The recieved and corrected curve is shown in figure XXX. With this curve it is possible to get the searched characteristics remanence and coercivity. Therefore it is required to place reference curves at the intercept point of the curve with the axes. The value of the intercept point with the Y-axes we get XXXX

and with the X-axes XXXX. The data is put in chart XXX. Because the effective range is too little to measure the saturation voltage in the next step I_{max} is increased to $I_{max} = 2,3\text{V}$ therefor the resistance has to be changed to $R = 0,2\Omega$. The resultant curve is shown in figure XXX.

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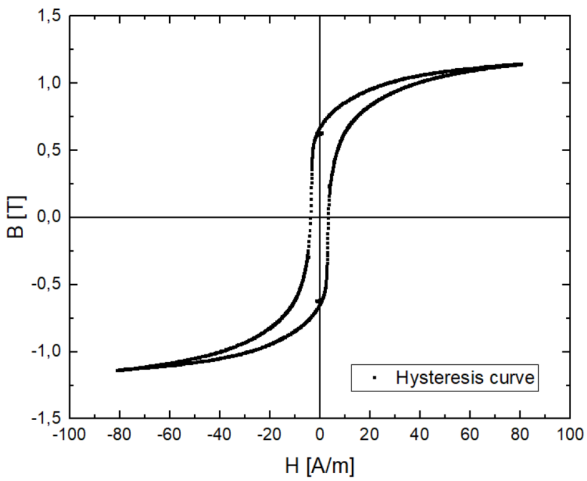
References regarding saturation magnetization and coercivity give the value of these characteristics for PERMENORM 5000H2 with $B_{saturation,Lit} = 1,55$ T and $H_{coercivity,Lit} = 5\ \frac{\text{A}}{m}$



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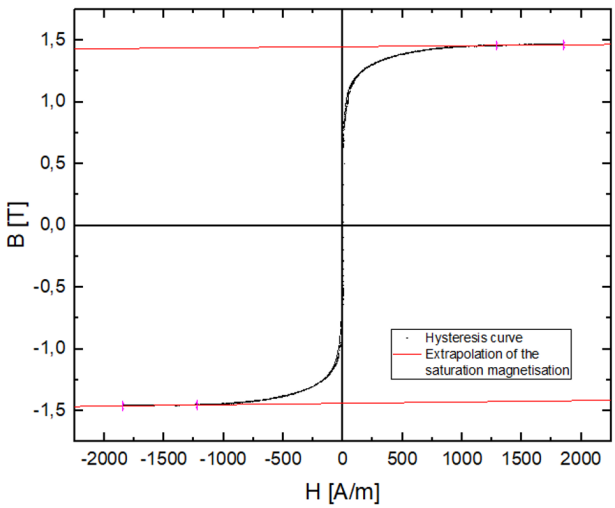

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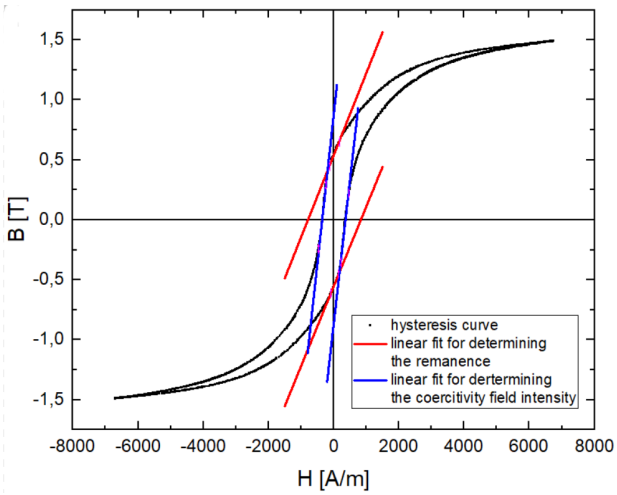
3.2 Flat steel type ST37K

In figure XXX the received hysteresis curve of flate steel is shown. The core of the coil is of the type ST37K. Again with the intercept points of curve and axes we get the values of remanence B_{rem} and coercivity $H_{coercivity}$. Because it was not possible to get the saturation magnetisation there is no value measured in this task. The core loss ϵ_{loss} in this experiment is calculated through numerical integration of the area in the curve with fomurla (1).

Additional to the normal hysteresis curve, in figure [xx] the change of direction of the graph is analysed. Therefore the normal procedure plotting the curve was done, but after reaching the starting point a second round was started. However in this following circle in two points the magnetic field was not constantly decreased but changed for a short time to go larger again with increasing H. So at two spots the field of H was shifted. In figure [xx] there are two little loops circulating anti- clockwise exact at the variations of the field. This effect shown can be used to demagnetise a magnet. This technique was also analysed and tested. The outcome of this demagnetisation is given in figure [xxx]. For this purpose a normal hysteresis curve was detected but bevor reaching the starting point the field H should already be decreased. This execution is repeated until the curve approximates zero. After this the outer magnetic field is increased until the starting point is hit again. So the resulting plot is a closed hysteresis curve with some loops inside. The actual execution was a little bit hard, hence a small remanence is remaining. But comparable with the earlier left remanence this one is way smaller which is indicating that this method works very well.



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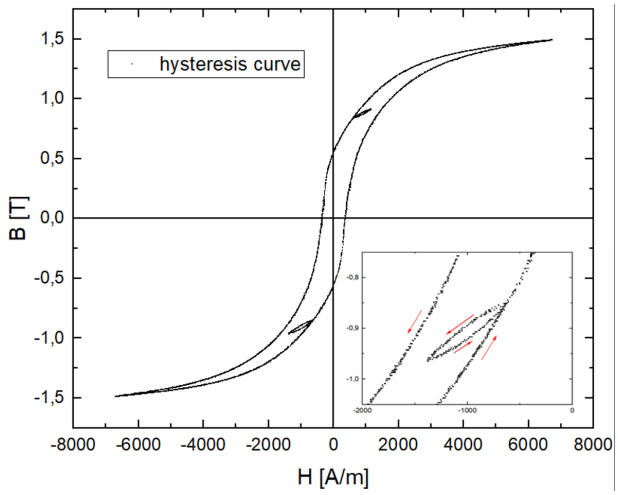
3.3 5000Z

PERMENORM 5000Z is another iron- nickel alloying used on material. The instruction to get the characteristics is identical to the other core with an alloying. The values of remanence and saturation magnetisati-



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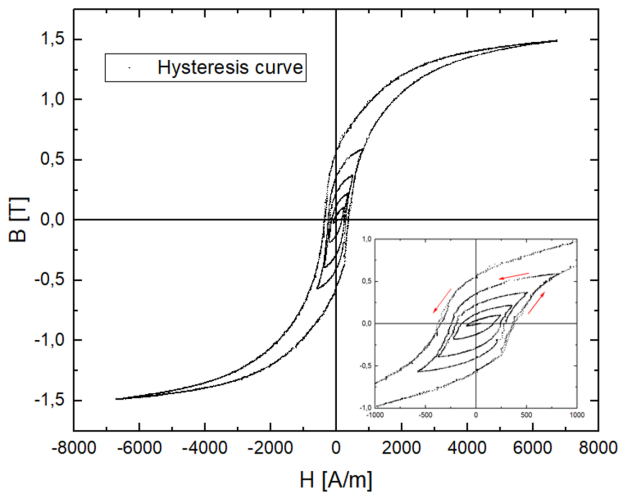

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on are given in table [xx]. Because there is no given data for the density of this material it is not possible to calculate ϵ_{loss}.

!!!!Verwendung und Einordnung des Materials in Weich/Hart Magnete anhand Koerzitivfeldstärke und sowas!!!!

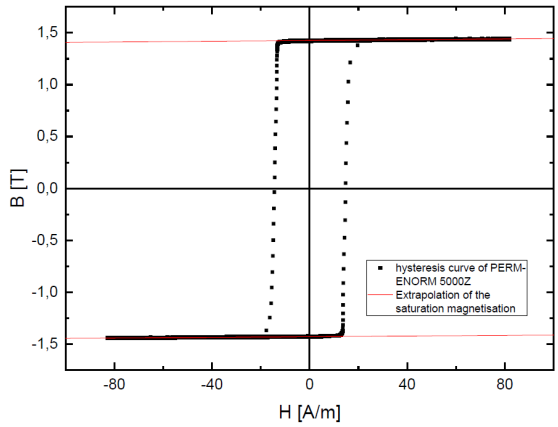


Abb. 7

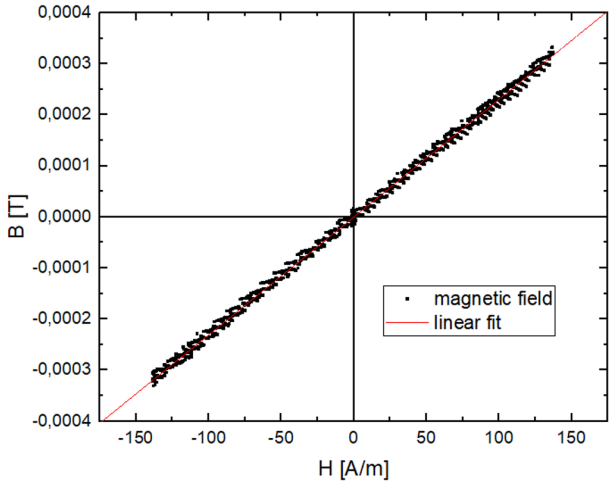
3.4 wooden core for comparison

The course of action is in these cases always the same. So the hysteresis curve of wood is plotted and shown in figure [xx]. This material was selected to be able to compare a ferromagnetic substance with a non ferromagnetic one. In the image the linear correlation of B and H is nicely illustrated. Also it is recognisable that the inner magnetic field is diminutive because wood is not magnetic so there is no inducting field possible. This is due to the inner structure of wood. The connection of the atom spin do not connect with each other as strong as in ferromagnets so it is impossible to bulid up magnetic domains. Because these bonds are so weak the termic fluctation after decreasing the outer field is relatively strong and destroys any parallelism. Therefore there is no hysteresis curve nor a left remanence. The given gradient of the curve is exactly the permeability $\mu = \mu_0 * \mu_r$.

!!!! Ergebnisse der steigung, einordnung des Holzes und vergleich mit literaturwert!!!!!

4 Results

In this paper three different ferromagnets and one theoretical diamagnet were analysed. With the help of plotted hysteresis curves it was possible to calculate characteristics like remanence, saturation magnetisation and some other ones which are all specific for those materials. To be able to compare the behavior of ferromagnets with non magnetical substances



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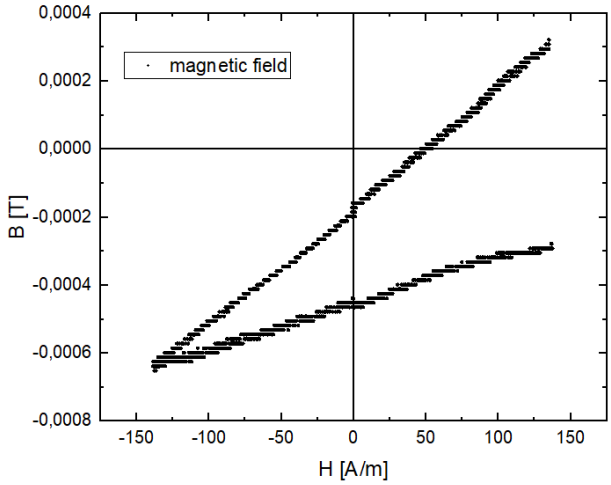

Abb. 9

wood was the material of choice. Now all values of the individual cores are listed beneath.

!!! Alle Werte nochmal notieren!!!

All curves are corresponding to the theoretical expectations. Additional the turnarround in the hysteresis curve was detected with the steel ST37K core. With this technique it was possible to demagnetize the mentioned core and to plot the received curve.

In Gl. (**??**) und Gl. (**??**)

FF: Angabe der verwendeten Literatur mit Quellennachweis.

Literatur

[1] Meschede, Gerthsen Physik, 23, 2005

[2] Frauenfelder und Huber, Einführung in die Physik II,

[3] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 39

[4] https://vacuumschmelze.com/products/soft-magnetic-materials-and-stamped-parts/40-to-50-Nickel-Iron

Aufgerufen am: 27.03.2022 um 15:42 Uhr

5 Anhang

5.1 Drift of the fluxmeter

In figure 9 the unadjusted plot of the wooden core is represented. The B- offset is aligned but not the drift of the fluxmeter. As you can probably tell the flux density is drifting towards higher values with a seemingly constant speed. Hence the final plot is not closed and zero is not hit. With the following formula it is possible to calculate the drift of the fluxmeter for each datapoint and hereby adjust the plot correctly.

$$\dot{B}_{drift} = (\frac{\partial B}{\partial t})_{drift} \approx \frac{B_n - B_1}{t_{total}} \tag{6}$$

With the use of the first recorded flux density B_1 and the last one B_n and the duration of the whole testseries t_{total}. So this received time shift was added to all datapoints to get the adjusted final plot.

5.2 troubleshooting

The given values at the workplace were supposed to be accurate and without error. So it is possible to calculate σ_B and σ_H with gaussian error propagation and the formulas [2] and [3].

The error of the fluxmeter is estimatet because the manufacturer information were missing. We set 0,5as the value of the error. For the other two used devices Voltcraft M3610D and the interface we took the errors of the manufacturer's data $\sigma_{U,interface} = 1,22$ m$\overline{V}$.